WORKED SOLUTION

STEP II 1991, Question 1

Pure • Quadratics and Discriminants

STEP II 1991, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q1

The key idea

A quadratic is a constant multiple of a perfect square exactly when it has a repeated root, that is, when its discriminant is zero. Every later part is that one condition applied to a quadratic whose coefficients depend on λ. Asking for the λ that make the discriminant vanish gives a quadratic *in* λ, and the decomposition then comes from solving two linear equations for f and g.

Solution

The condition for the form $(*)$

Expanding,

$$a(x+k)^2 = ax^2 + 2akx + ak^2.$$

If $\mathrm{h}(x)$ has this form then, comparing coefficients, $b = 2ak$ and $c = ak^2$, so

$$b^2 = 4a^2k^2 = 4a \cdot ak^2 = 4ac.$$

Conversely, if $b^2 = 4ac$, take $k = \dfrac{b}{2a}$ (allowed since $a \neq 0$). Then $2ak = b$ and $ak^2 = \dfrac{b^2}{4a} = c$, so $\mathrm{h}(x) = a(x+k)^2$. The condition is therefore

$$b^2 = 4ac,$$

which says that the discriminant is zero.

The two values of λ

We have

$$\mathrm{f}(x) + \lambda \mathrm{g}(x) = (3+\lambda)x^2 + 4x - 2\lambda.$$

If $\lambda = -3$ this is $4x + 6$. That is not of the form $(*)$, which needs a nonzero coefficient of x^2. So $3 + \lambda \neq 0$, and by the condition above we need

$$16 = 4(3+\lambda)(-2\lambda) \iff \lambda^2 + 3\lambda + 2 = 0 \iff (\lambda+1)(\lambda+2) = 0.$$

So $\lambda = -1$ or $\lambda = -2$, and neither is -3. Explicitly,

$$f(x) - g(x) = 2x^2 + 4x + 2 = 2(x+1)^2,$$
$$f(x) - 2g(x) = x^2 + 4x + 4 = (x+2)^2.$$

The decomposition

Ask yourself How do two facts about $f - g$ and $f - 2g$ tell us f and g separately?

Treat the two displayed identities as simultaneous linear equations in the unknowns $f(x)$ and $g(x)$. Subtracting the second from the first gives

$$g(x) = 2(x+1)^2 - (x+2)^2,$$

and then $f(x) = (f(x) - 2g(x)) + 2g(x) = (x+2)^2 + 4(x+1)^2 - 2(x+2)^2$, that is,

$$f(x) = 4(x+1)^2 - (x+2)^2.$$

Checking by expansion: $4(x^2 + 2x + 1) - (x^2 + 4x + 4) = 3x^2 + 4x$, and $2(x^2 + 2x + 1) - (x^2 + 4x + 4) = x^2 - 2$. So one choice is

$$A = 4, \quad B = -1, \quad C = 2, \quad D = -1, \quad m = 1, \quad n = 2.$$

Swapping the roles of m and n gives the equivalent answer $m = 2$, $n = 1$, $A = -1$, $B = 4$, $C = -1$, $D = 2$.

The final part

Now

$$f(x) + \lambda g(x) = (3 + \lambda)x^2 + 4x + \lambda\alpha.$$

As before, $\lambda = -3$ gives the linear expression $4x - 3\alpha$, which is never of the form $(*)$. For $\lambda \neq -3$ the condition $b^2 = 4ac$ reads

$$16 = 4(3+\lambda)\lambda\alpha \iff \alpha\lambda^2 + 3\alpha\lambda - 4 = 0. \qquad (\dagger)$$

We want exactly one value of λ satisfying $(\dagger)$ with $\lambda \neq -3$.

Ask yourself Could $(\dagger)$ have two roots, one of which is the forbidden value $\lambda = -3$?

No. Putting $\lambda = -3$ into the left-hand side of $(\dagger)$ gives $9\alpha - 9\alpha - 4 = -4 \neq 0$, so $\lambda = -3$ is never a root. The values of λ that work are therefore exactly the roots of $(\dagger)$.

- If $\alpha = 0$, $(\dagger)$ becomes $-4 = 0$, which has no solutions. So $\alpha \neq 0$.
- If $\alpha \neq 0$, $(\dagger)$ is a genuine quadratic in λ. It has exactly one root precisely when its discriminant is zero:

  $$9\alpha^2 + 16\alpha = 0 \iff \alpha(9\alpha + 16) = 0 \iff \alpha = -\tfrac{16}{9},$$

  using $\alpha \neq 0$.

Hence

$$\alpha = -\frac{16}{9}.$$

The repeated root is $\lambda = -\frac{3\alpha}{2\alpha} = -\frac{3}{2}$, and indeed

$$f(x) - \tfrac{3}{2}\left(x^2 - \tfrac{16}{9}\right) = \tfrac{3}{2}x^2 + 4x + \tfrac{8}{3} = \tfrac{3}{2}\left(x + \tfrac{4}{3}\right)^2.$$

Checks

Expand the decomposition at a test value: at $x = 1$, $4(2)^2 - (3)^2 = 7 = f(1)$ and $2(2)^2 - 3^2 = -1 = g(1)$. In the final part, $\frac{3}{2}(x + \frac{4}{3})^2$ at $x = 0$ is $\frac{3}{2} \cdot \frac{16}{9} = \frac{8}{3}$, matching $\lambda\alpha = (-\frac{3}{2})(-\frac{16}{9})$. A computer testing every rational $\lambda = p/q$ with $|p| \leq 60$, $q \leq 12$ found exactly $\lambda = -1, -2$ in the middle part and only $\lambda = -\frac{3}{2}$ when $\alpha = -\frac{16}{9}$.

Common slips

- Forgetting the requirement $a \neq 0$ in $(*)$: at $\lambda = -3$ the expression is linear, and you should say why that value is excluded.
- In the final part, stopping at "discriminant of $(\dagger)$ is zero" without dealing with $\alpha = 0$, where $(\dagger)$ is not a quadratic at all and has no roots.

Beyond STEP

Writing f and g as combinations of two squares of linear forms is simultaneous diagonalisation of a pair of quadratic forms. The values of λ that make $f + \lambda g$ degenerate are the roots of $\det(F + \lambda G) = 0$ for the corresponding symmetric matrices, a "generalised eigenvalue" problem. The final part, where the two values of λ coincide, is exactly the case in which such a diagonalisation can fail. Strang's *Introduction to Linear Algebra* treats quadratic forms and eigenvalues in this spirit.

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